Reduced-unit planar Hooke impact

LAMMPS dem01-hooke-2d-lj | Automatic normal damping | 29 October 2026

PASS	All four runs pass official 2% restitution and momentum checks.

Official rebound checks at reduced time 0.1447

Run	Measured e	Error (%)	Left vx (m/s)	Right vx (m/s)	Official
CPU dt	0.5004258213	0.085164	-0.500425821	0.500425821	PASS
GPU dt	0.5004258213	0.085164	-0.500425821	0.500425821	PASS
CPU dt/2	0.5014963946	0.299279	-0.501496395	0.501496395	PASS
GPU dt/2	0.5014963946	0.299279	-0.501496395	0.501496395	PASS

Target e = 0.5; relative tolerance = 2%. The simultaneous momentum bound is abs(m v_left + m v_right) <= 0.02 (2m) vin = 0.02094395102 kg m/s. Both checks apply at step 1447. Pinned saved-state e = 0.500425821332 is a supplemental reference.

Official inputs and reduced-unit realization

Quantity	Unchanged numeric input / native mapping
Unit scales	L0 = 1 m, M0 = 1 kg, T0 = 1 s; energy scale = 1 J. All official numbers remain unchanged. units lj supplies reduced units; pair_style granular supplies contact, with no LJ pair potential.
Particles / motion	R = 0.5 m; density = 1 kg/m^3; centers x = +/-0.55 m; incoming vx = +/-1 m/s. Zero transverse motion, initial spin, gravity and applied forces.
Contact / damping	Hooke kn = 10000 N/m; e = 0.5. Automatic cn = 22.04795889 N s/m; manual overrides off. All tangential and rotational terms are zero.
Time / full output	dt = 0.0001 / 0.00005 s; 1447 / 2894 steps to 0.1447 s. Every numerical step, including t = 0, is retained.

Mass matches; disc inertia is explicitly outside active scope

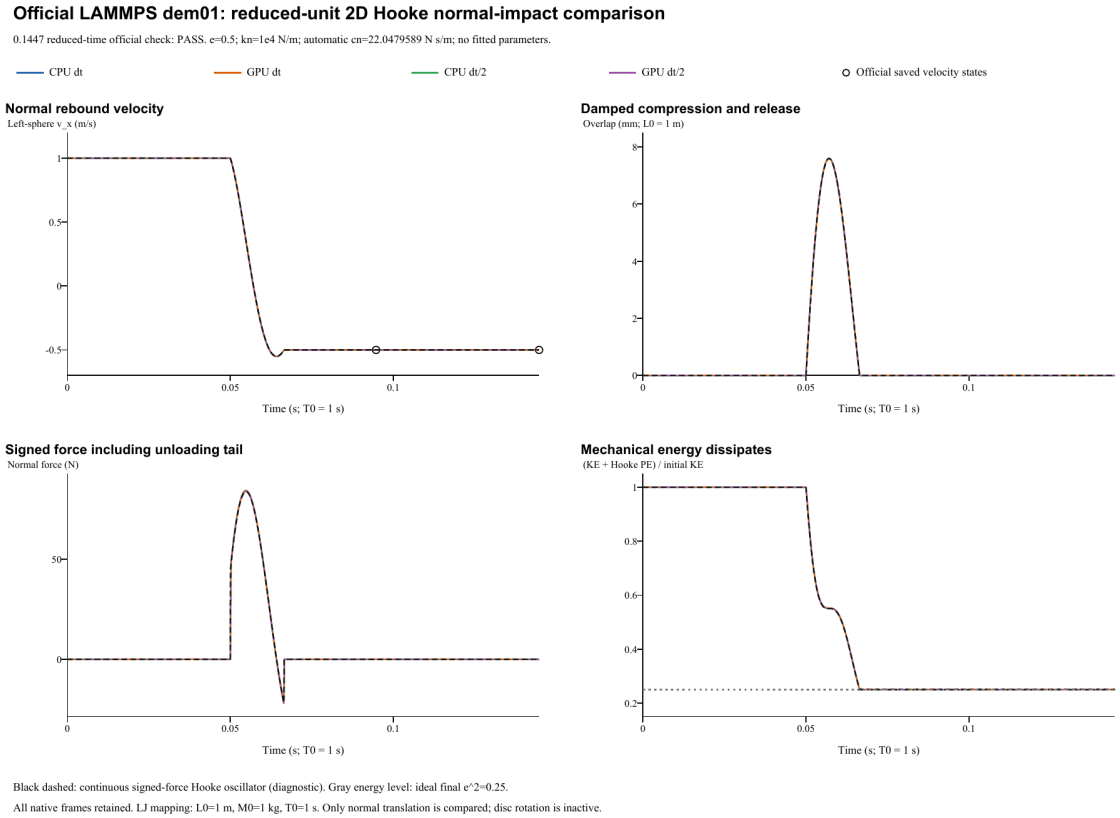
Pinned set.cpp uses sphere-volume mass for ordinary density, including dimension 2: m = 4 pi rho R^3/3 = pi/6 = 0.5235987756 kg. Only density/disc would change that mass rule. nve/sphere disc changes inertia to mR^2/2 = 0.06544984695 kg m^2; native sphere inertia is 2mR^2/5 = 0.05235987756 kg m^2.

The inertias differ but torque and spin stay zero in this exact central collision. The planar enforce2d subspace is invariant. This verifies active normal translation, not disc rotation, a general 2D solver or LJ forces. No density adjustment, parameter fitting or solver modification was made.

Normal law and reference provenance

cn = 2 beta sqrt(m* kn); beta = -ln(e)/sqrt(pi^2 + ln(e)^2); m* = m/2.
Fn = kn delta - cn vn_assembly. Signed viscous unloading is retained without force clipping.
Official input | Analytic checker | Mass/inertia/unit source audit. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob 9bc3f60cd4c5059210c74c24a8bb40a5b822ed17. LAMMPS was not rerun locally. YAML epsilon 1e-10 is a same-code regression tolerance.

Native histories and retained evidence



Automatic damping and dissipative response

Run	Inferred cn (N s/m)	Min. Fn (N)	Final KE / initial	Max. momentum
CPU dt	22.047958895	-21.974154	0.250426003	0
GPU dt	22.047958895	-21.974154	0.250426003	0
CPU dt/2	22.047958895	-21.760073	0.251498634	0
GPU dt/2	22.047958895	-21.760073	0.251498634	0

Inferred automatic cn agrees with its analytic value to maximum relative error 7.22e-15; maximum signed-force reconstruction residual is 1.42e-14 N. This is an inferred coefficient, not a direct native output channel.

CPU/GPU whole-history maximum velocity / signed-normal-force differences are 2.22e-16 m/s / 2.84e-14 N. Halving dt changes CPU restitution by 0.00107057. Relative restitution error is 0.085164% / 0.299279%; this refinement does not show monotonic improvement, while both remain below 2%.

Maximum linear impulse-balance residual is 9.99e-16 kg m/s. Mechanical energy dissipates; the ideal final kinetic-energy fraction is e^2 = 0.25.

Inference removes the last half-kick: vn_assembly = vn_saved - dt Fn/(2m*). The signed force and both particles' impulse balance are retained. Black dashed oscillator curves are supplemental; zero transverse motion, torque and spin keep the different inertias inactive.

Complete results and scope

1448 frames per original-step run; 2895 per half-step run: all 8686 native frames, contacts, projects, every-step CSVs, audit hashes and CPU/CUDA logs are retained. All four runs were newly executed.

Run manifest | Analysis JSON | Checkpoints | Full native ZIP | Retention hashes

Real SimulationSession uses linked FunDEM 0.4.3 CPU/CUDA libraries; installed GUI byte identity is not established. Scope: collinear normal Hooke translation, reduced-unit realization, automatic normal damping, signed unloading, momentum and dissipation. Active disc rotation, general 2D motion, friction/history and LJ pair forces remain outside scope.